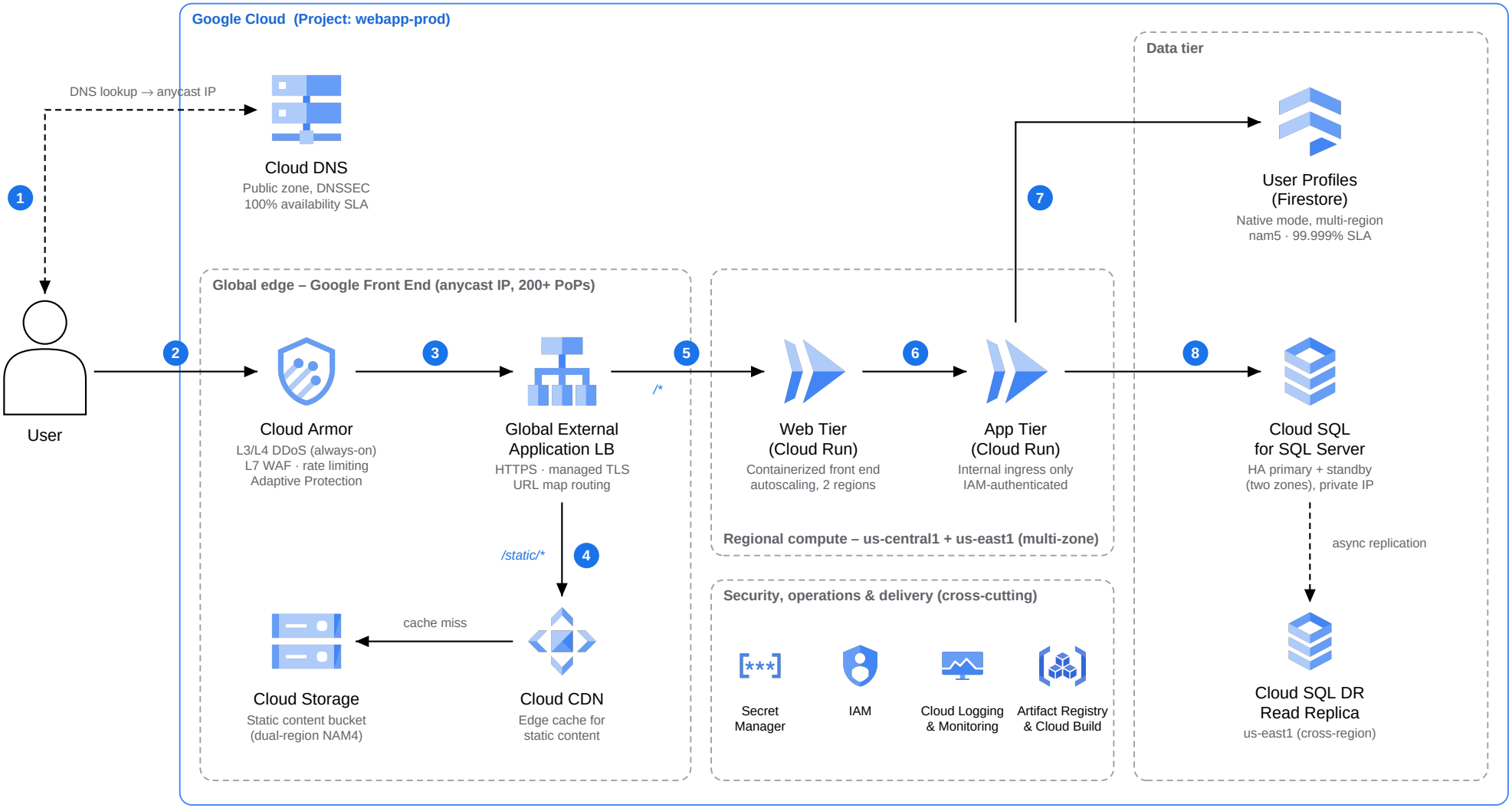
Proposed Architecture – Google Cloud (modernized web application)

Target-state data flow for the current on-premises application (User → NGFW → DNS → Load Balancer → Web Server → App Server → Redis / SQL Server).



Legend & notes

Request / data flow — DNS resolution / replication — 1 Step number – matches the bullet order in Proposed_Architecture_Summary.docx

- Responses return along the same path (App Tier → Web Tier → Load Balancer → User). Cacheable responses are cached by Cloud CDN at the edge.
- Cloud Armor replaces the on-prem NGFW at the internet edge; Cloud DNS replaces the DNS server; Global External Application LB replaces the on-prem load balancer.
- Static files move from the web server to Cloud Storage + Cloud CDN. Web/App servers are containerized on Cloud Run. Redis user profiles/activity move to Firestore; SQL Server moves to Cloud SQL for SQL Server.
- Web and App tiers run active-active in us-central1 and us-east1 behind the single global anycast IP (serverless NEGs); the LB sends users to the closest healthy region.
- Cloud SQL and Cloud Run have no public exposure: Cloud SQL uses private IP only (Private Service Access); Cloud Run ingress is restricted to the LB (web) or internal VPC traffic (app).
- Icons: official Google Cloud icon library (cloud.google.com/icons). Region names are examples and will be confirmed with the customer.